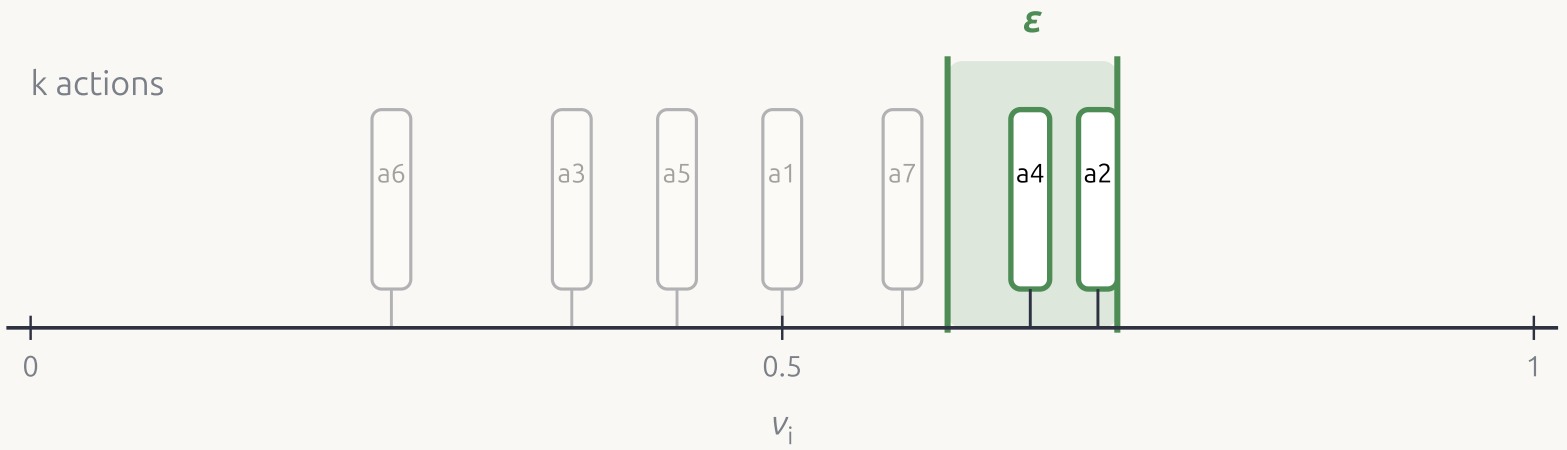
Coherent Rollout Oracles for Finite-Horizon Sequential Decision Problems

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The Selection Problem



$$\Pr\left[v_{\hat{a}} \geq \max_j v_j - \varepsilon \right] \geq \frac{2}{3}$$

Speedup

Quantum

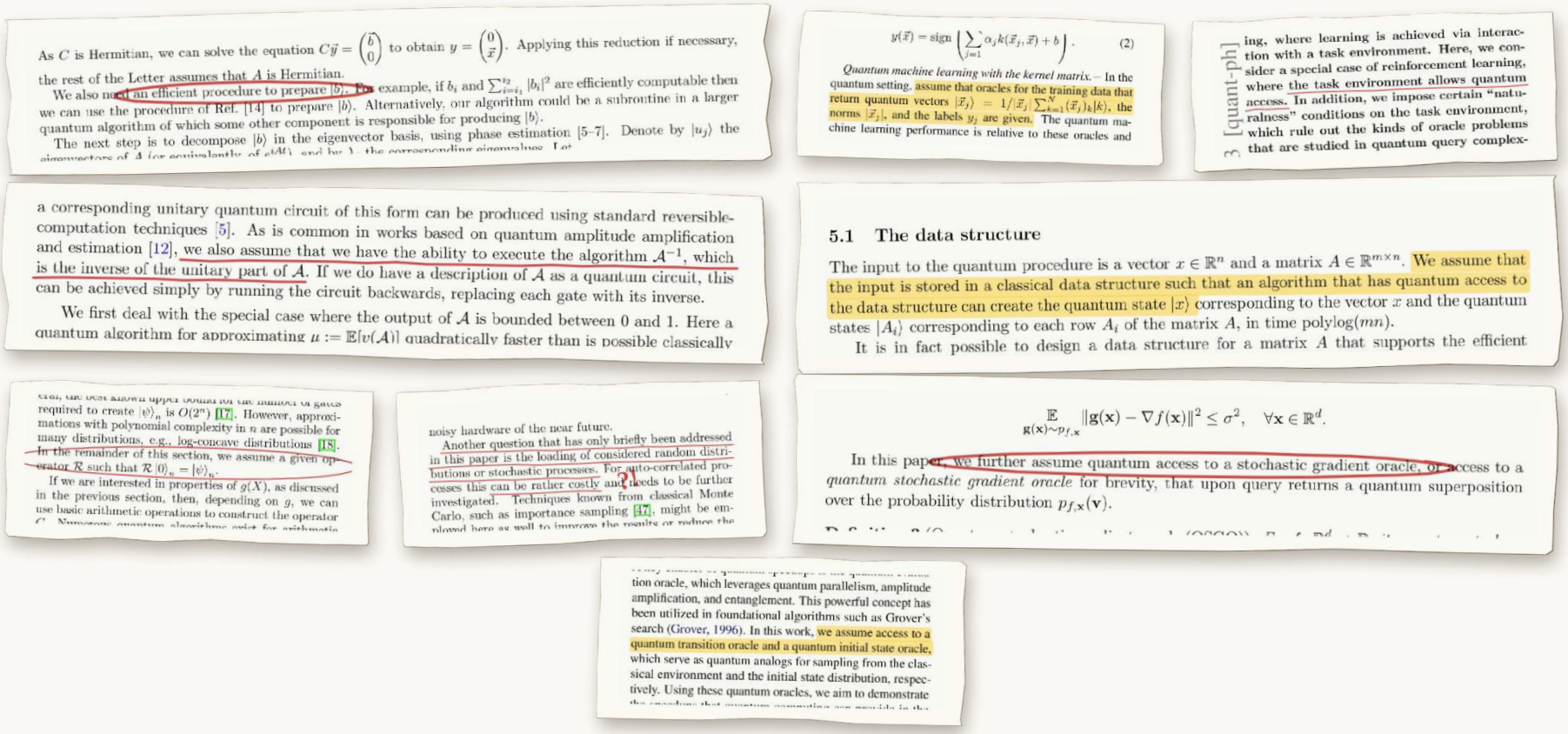
$\widetilde{O}(\sqrt{k}/\varepsilon)$

Classical

$\Omega(k/\varepsilon^2)$

Wang et al.

“Assume oracle access.”



As C is Hermitian, we can solve the equation $C\vec{y} = \begin{pmatrix} \vec{b} \\ 0 \end{pmatrix}$ to obtain $y = \begin{pmatrix} 0 \\ \vec{x} \end{pmatrix}$. Applying this reduction if necessary, the rest of the Letter assumes that A is Hermitian.

We also need an efficient procedure to prepare $|b\rangle$. For example, if b_i and $\sum_{i=i_1}^{i_2} |b_i|^2$ are efficiently computable then we can use the procedure of Ref. [14] to prepare $|b\rangle$. Alternatively, our algorithm could be a subroutine in a larger quantum algorithm of which some other component is responsible for producing $|b\rangle$.

The next step is to decompose $|b\rangle$ in the eigenvector basis, using phase estimation [5–7]. Denote by $|u_j\rangle$ the

$$y(\vec{x}) = \text{sign}\left(\sum_{j=1}^{} \alpha_j k(\vec{x}_j, \vec{x}) + b\right). \quad (2)$$

Quantum machine learning with the kernel matrix. – In the quantum setting, assume that oracles for the training data that return quantum vectors $|\vec{x}_j\rangle = 1/|\vec{x}_j| \sum_{k=1}^{N} (\vec{x}_j)_k |k\rangle$, the norms $|\vec{x}_j|$, and the labels y_j are given. The quantum machine learning performance is relative to these oracles and

3 [quant-ph]

ing, where learning is achieved via interaction with a task environment. Here, we consider a special case of reinforcement learning, where the task environment allows quantum access. In addition, we impose certain “naturalness” conditions on the task environment, which rule out the kinds of oracle problems that are studied in quantum query complex-

a corresponding unitary quantum circuit of this form can be produced using standard reversible-computation techniques [5]. As is common in works based on quantum amplitude amplification and estimation [12], we also assume that we have the ability to execute the algorithm $\mathcal{A}^{-1}$, which is the inverse of the unitary part of $\mathcal{A}$. If we do have a description of $\mathcal{A}$ as a quantum circuit, this can be achieved simply by running the circuit backwards, replacing each gate with its inverse.

We first deal with the special case where the output of $\mathcal{A}$ is bounded between 0 and 1. Here a quantum algorithm for approximating $u := \mathbb{E}[v(\mathcal{A})]$ quadratically faster than is possible classically

5.1 The data structure

The input to the quantum procedure is a vector $x \in \mathbb{R}^n$ and a matrix $A \in \mathbb{R}^{m \times n}$. We assume that the input is stored in a classical data structure such that an algorithm that has quantum access to the data structure can create the quantum state $|x\rangle$ corresponding to the vector x and the quantum states $|A_i\rangle$ corresponding to each row A_i of the matrix A, in time polylog(mn).

It is in fact possible to design a data structure for a matrix A that supports the efficient

ural, the best known upper bound for the number of gates required to create $|\psi\rangle_n$ is $O(2^n)$ [17]. However, approximations with polynomial complexity in n are possible for many distributions, e.g., log-concave distributions [18]. In the remainder of this section, we assume a given operator $\mathcal{R}$ such that $\mathcal{R}|0\rangle_n = |\psi\rangle_n$.

If we are interested in properties of $g(X)$, as discussed in the previous section, then, depending on g, we can use basic arithmetic operations to construct the operator

noisy hardware of the near future.

Another question that has only briefly been addressed in this paper is the loading of considered random distributions or stochastic processes. For auto-correlated processes this can be rather costly and needs to be further investigated. Techniques known from classical Monte Carlo, such as importance sampling [47], might be employed here as well to improve the results or reduce the

$$\mathop{\mathbb{E}}_{\mathbf{g}(\mathbf{x}) \sim p_{f,\mathbf{x}}} \|\mathbf{g}(\mathbf{x}) - \nabla f(\mathbf{x})\|^2 \leq \sigma^2, \quad \forall \mathbf{x} \in \mathbb{R}^d.$$

In this paper, we further assume quantum access to a stochastic gradient oracle, or access to a *quantum stochastic gradient oracle* for brevity, that upon query returns a quantum superposition over the probability distribution $p_{f,\mathbf{x}}(\mathbf{v})$.

tion oracle, which leverages quantum parallelism, amplitude amplification, and entanglement. This powerful concept has been utilized in foundational algorithms such as Grover's search (Grover, 1996). In this work, we assume access to a quantum transition oracle and a quantum initial state oracle, which serve as quantum analogs for sampling from the classical environment and the initial state distribution, respectively. Using these quantum oracles, we aim to demonstrate

Obstacle

- Randomness in explicit registers
- Unitary, invertible rollout step
- Selector to valid action, without measurement or rejection
- Branch-dependent valid actions
- Sentinel when the rank is out of range

Contribution 1

A coherent rank-select decoder for branch-dependent valid actions

- We give two coherent rank-select constructions, both using $O(w)$ clean ancillae: a sequential scan with $O(Nw)$ gates that is optimal in the bounded-span model, and a blocked construction with $O(N \log w)$ gates that exploits long-range connectivity.
- The unconditional gate lower bound across all layouts is $\Omega(N)$.
- To our knowledge, this is the first reversible-circuit complexity analysis of coherent rank-select under branch-dependent validity.

Contribution 2

An explicit polynomial-size coherent rollout oracle for implicit-state finite-horizon planning

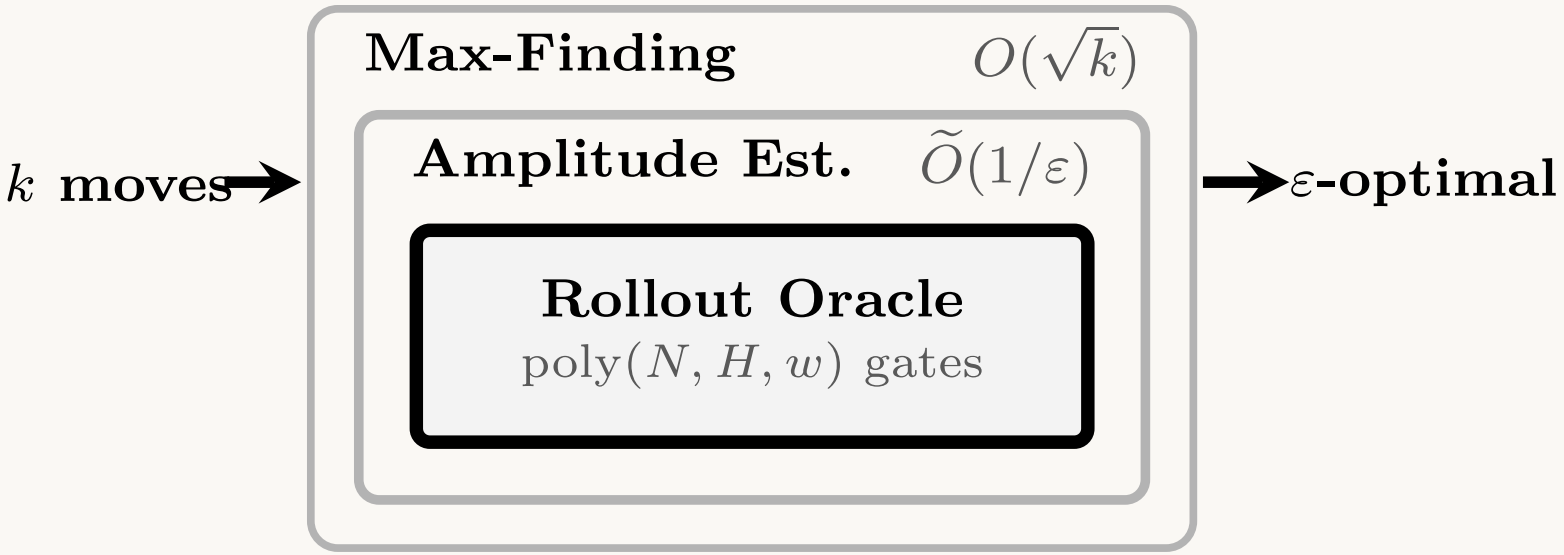
- We decompose the oracle into three reversible phases (rank-select indexing, stochastic transition, terminal evaluation) that compose into a polynomial-size unitary, with gate-count and qubit-count formulas per call.
- Plugged into the pipeline of Wang et al., the oracle yields an explicit rollout algorithm using $\widetilde{O}(\sqrt{k}/\varepsilon)$ coherent calls.
- It is implemented in Qiskit with branchwise agreement against classical rollouts on all tested instances.

Contribution 3

Lower-bound robustness across exponentially many configurations

- A bounded-influence lifting theorem and a per-configuration transfer theorem carry the classical $\Omega(k/\varepsilon^2)$ hard construction from a base configuration to every member of an exponential family of locally coupled configurations.

Preliminaries

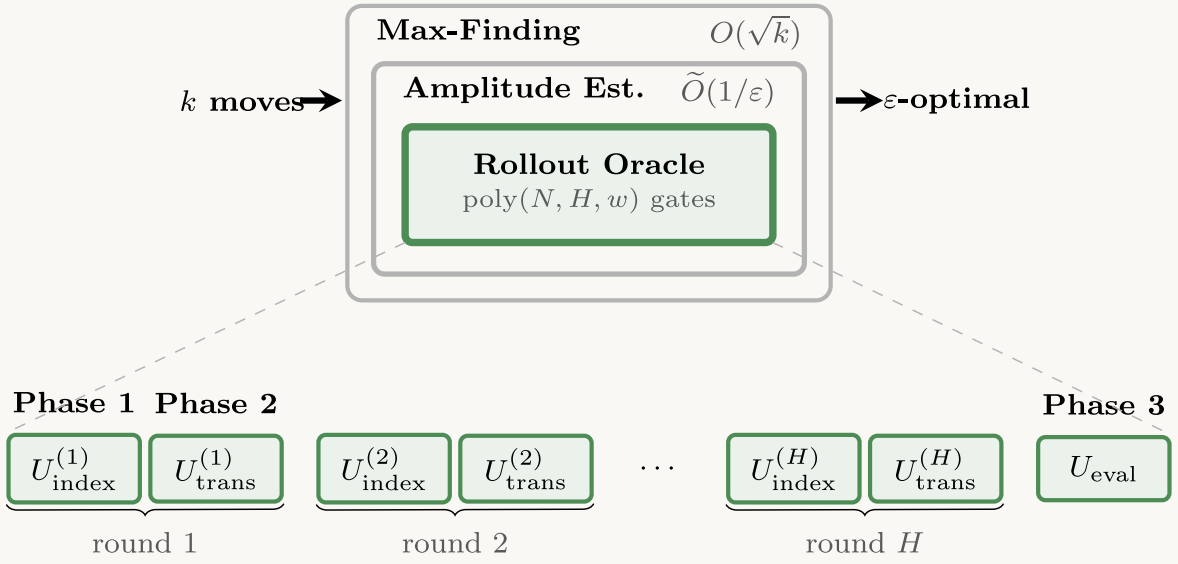


Recurring notation

k	arms (first actions)	v_i	value of action i
ε	accuracy target	H	rollout horizon
N	positions per round	w	$\lceil \log_2(N+1) \rceil$
M_s	validity mask in state s	r	rank into M_s
$C^*,\ a^*$	witness config., action	$\mathcal{F}$	peripheral family
Q, δ_u	peripheral set; influence	β	$\sum_{u \in Q} \delta_u$

A Normal Form for Coherent Rollout Oracles

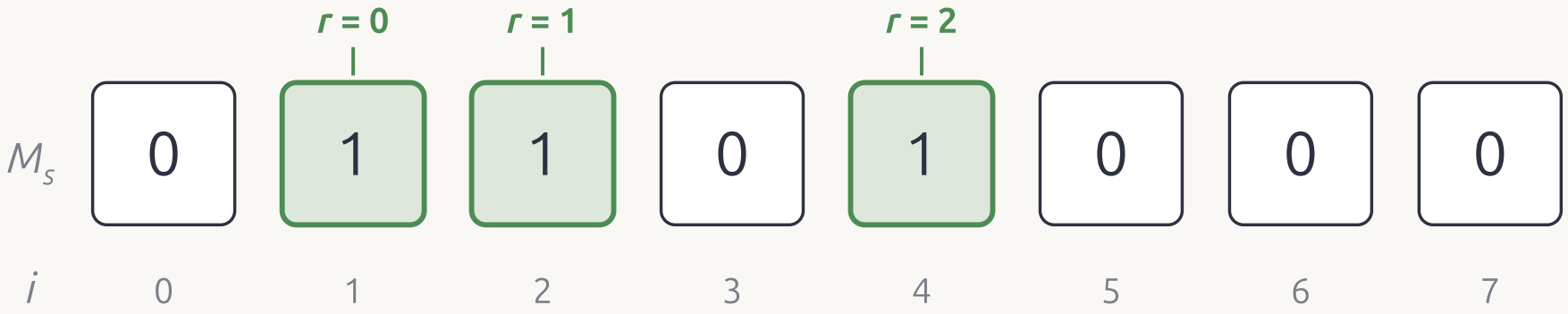
Three reversible phases



$$U \;=\; U_{\text{eval}} \cdot \prod_{h=H}^{1} \left(U^{(h)}_{\text{trans}} \cdot U^{(h)}_{\text{index}} \right)$$

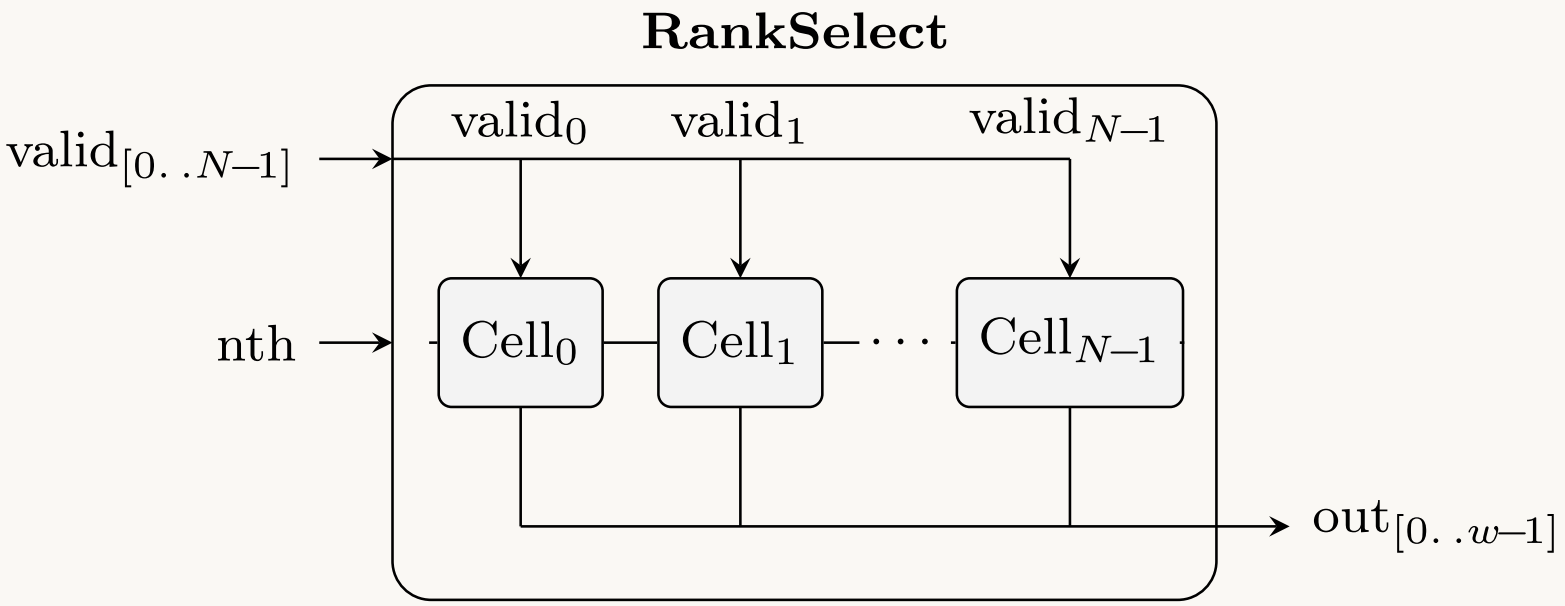
Phase 1: Coherent Rank-Select Indexing

$$|s\rangle|r\rangle|0\rangle^{\otimes w} \;\mapsto\; |s\rangle|r\rangle|j_s(r)\rangle$$

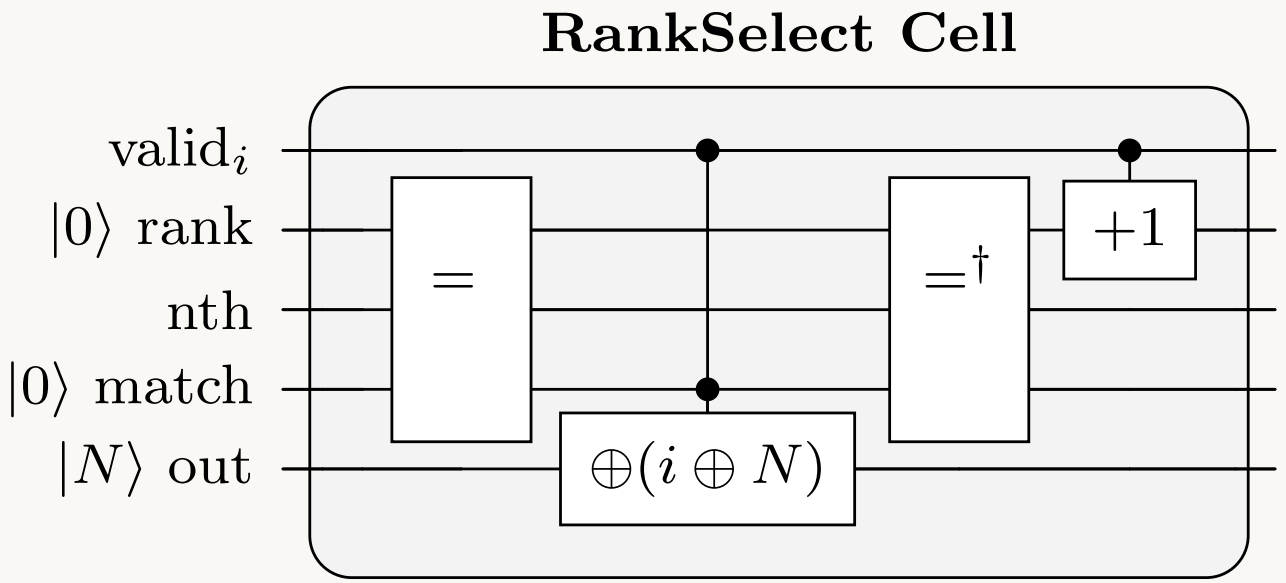


$$j_s(0) = 1, \quad j_s(1) = 2, \quad j_s(2) = 4, \quad j_s(r) = 8 \text{ for } r \geq 3$$

RankSelect


RankSelect
valid$_{[0..N-1]}$
nth
valid$_0$
valid$_1$
valid$_{N-1}$
Cell$_0$
Cell$_1$
Cell$_{N-1}$
out$_{[0..w-1]}$

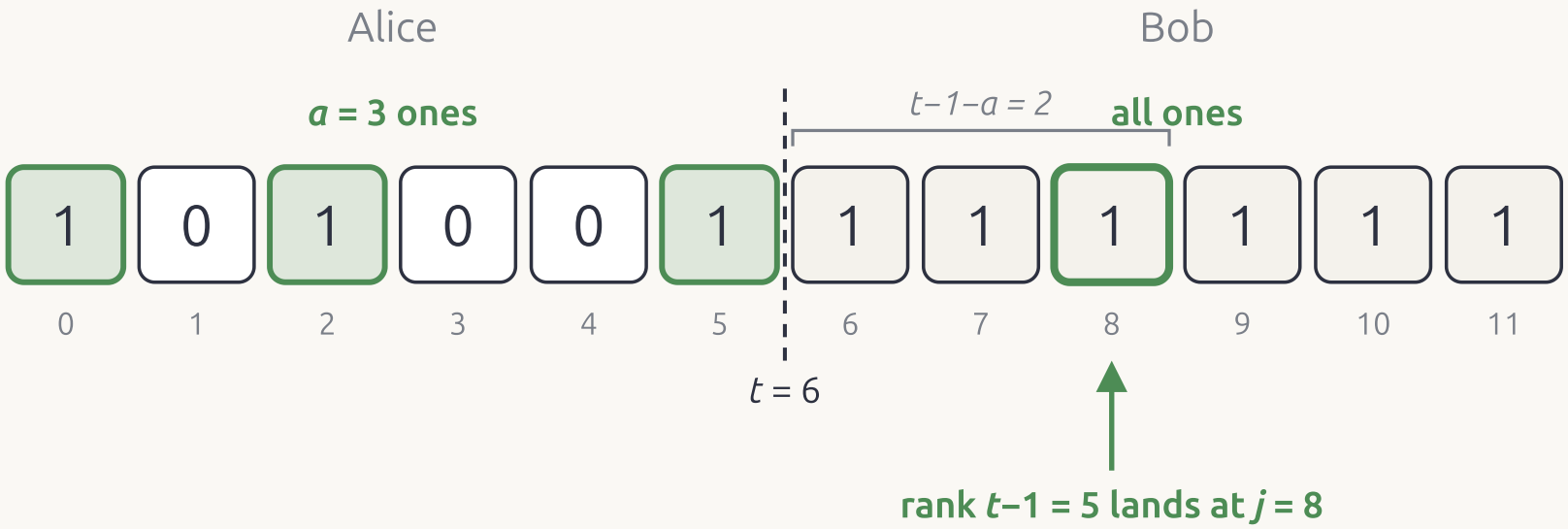
RankSelect Cell



Gate bounds

	Gates	**Ancillae**	**Model**
Sequential scan	$O(Nw)$	$O(w)$	bounded-span, gate-optimal
Blocked	$O(N \log w)$	$O(w)$	long-range gates
Any circuit	$\Omega(N)$		bounded fan-in

Rank-select cut lower bound



$$j = t + (t{-}1{-}a) \quad \Rightarrow \quad \lceil \log_2 \min(t, N-t) \rceil \text{ bits across cut } t$$

Phase 2: Reversible Stochastic Transition



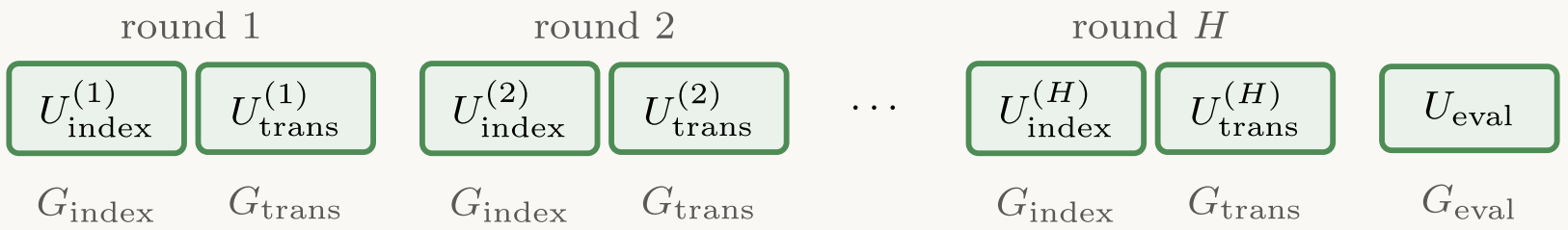
$O(1)$ per cell, $O(N)$ per round

Phase 3: Coherent Terminal Evaluation

$$\Pr[\text{payoff} = 1] = v_i$$

- Local predicates: $O(N)$
- Comparing two w-bit counts: $O(N^2 w)$

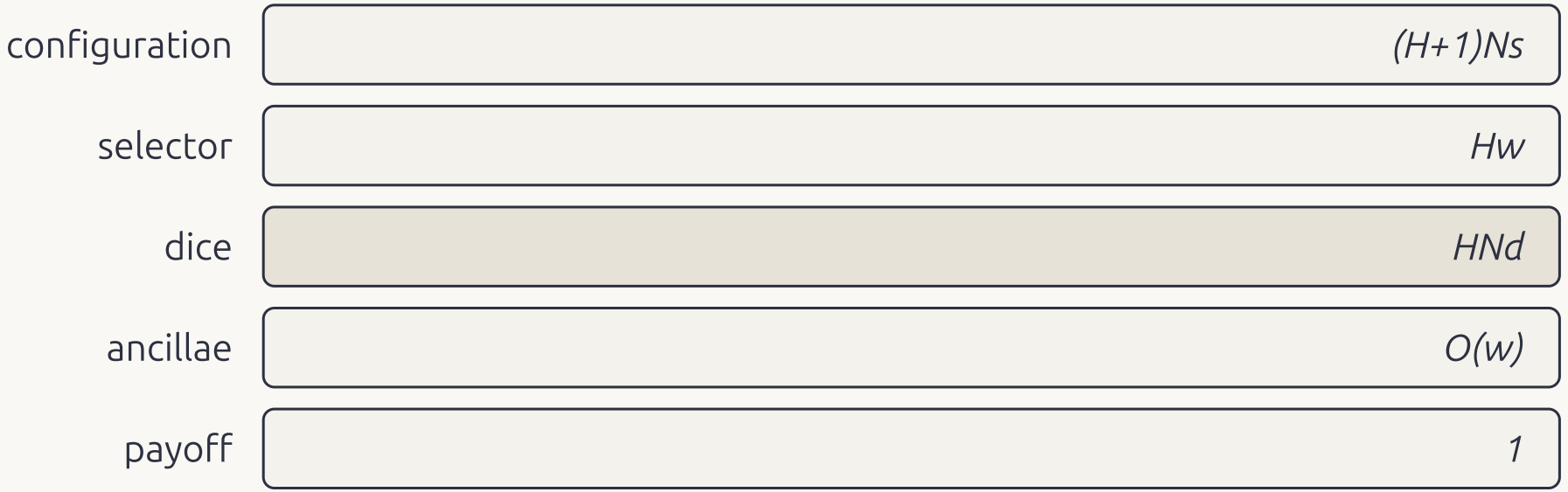
Per-Call Gates



$$G_{\text{call}} = H\,(G_{\text{index}} + G_{\text{trans}}) + G_{\text{eval}}$$

Bounded-span $O(HNw + N^2w)$, long-range $O(HN\log w + N^2w)$

Per-Call Qubits

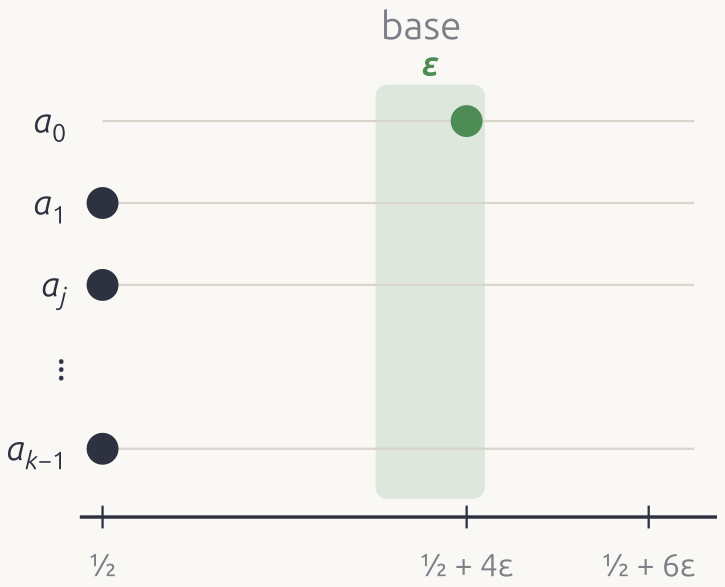


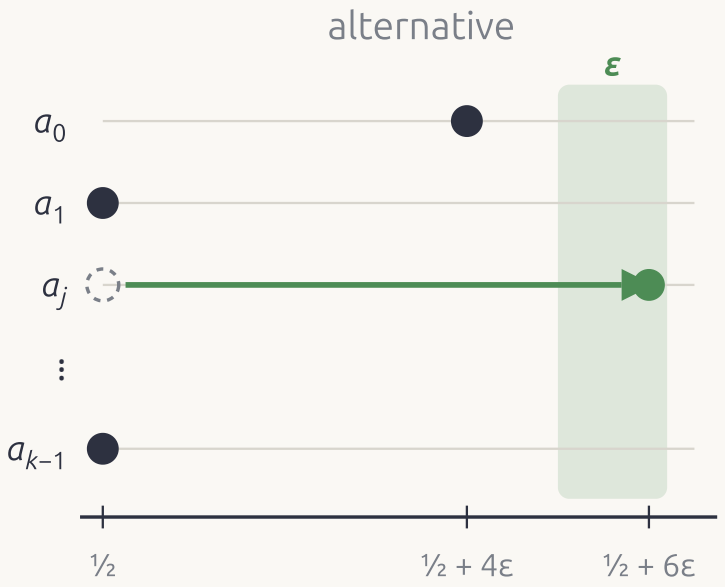
$$Q_{\text{call}} = (H+1)Ns + Hw + HNd + q_{\text{anc}}$$

s configuration state width, d dice width

Classical Lower Bound

Base and alternative





Change of measure

Information needed

$$\mathrm{kl}\left(\tfrac{1}{3}, \tfrac{2}{3}\right) = \frac{\ln 2}{3}$$

Information per rollout

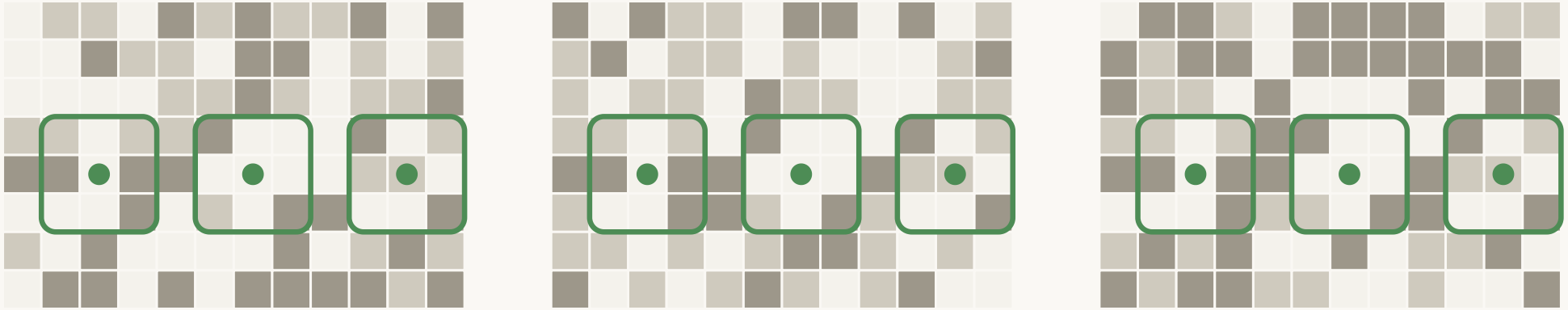
$$\mathrm{kl}\left(\tfrac{1}{2}, \tfrac{1}{2} + 6\varepsilon\right) \leq 96\,\varepsilon^2$$

$$\mathbb{E}_0[N_j] \;\geq\; \frac{\ln 2/3}{96\,\varepsilon^2} \;=\; \frac{\ln 2}{288\,\varepsilon^2}$$

$$\mathbb{E}_0[\tau] \;\geq\; \frac{(k-1)\ln 2}{288\,\varepsilon^2} \;=\; \Omega\left(\frac{k}{\varepsilon^2}\right)$$

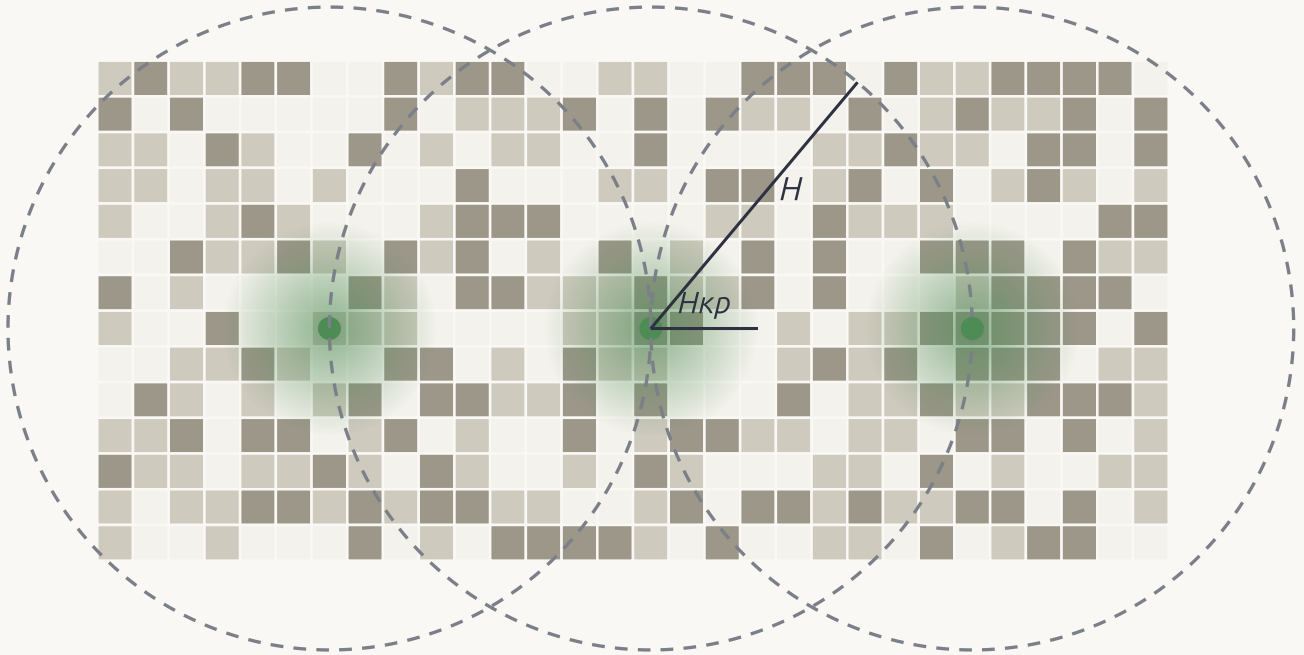
Kaufmann, Cappé, Garivier 2016 · Lattimore, Szepesvári 2020

Bounded-influence lifting



$$\mathbb{E}_C[\tau] \;\geq\; \frac{(k-1)\,\ln 2}{288\,\varepsilon^2} \;=\; \Omega\!\left(\frac{k}{\varepsilon^2}\right) \qquad \text{for every } C \in \mathcal{F}$$

Subcritical influence decay

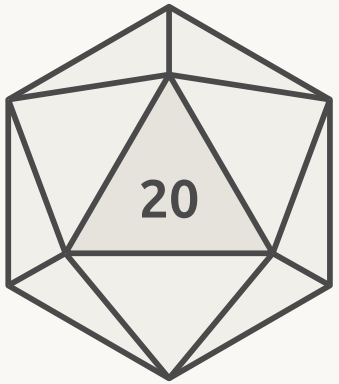


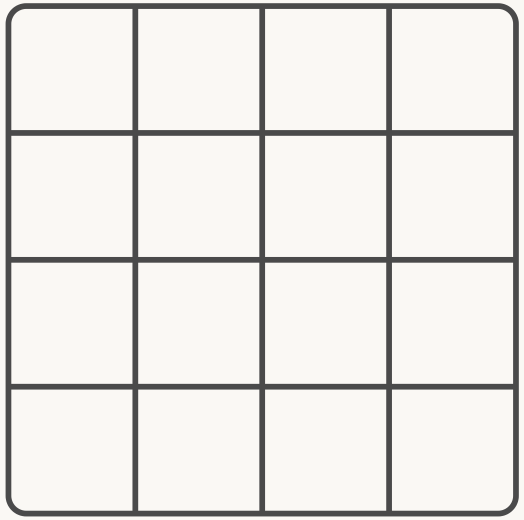
$$\delta(d) \;\leq\; \frac{\kappa}{\kappa - 1}\left((\kappa - 1)\,p\right)^{d} \qquad \kappa p < 1$$

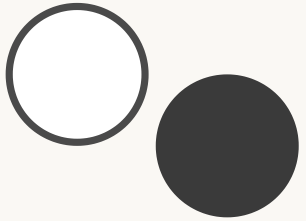
Quantum Upper Bound

Instantiation

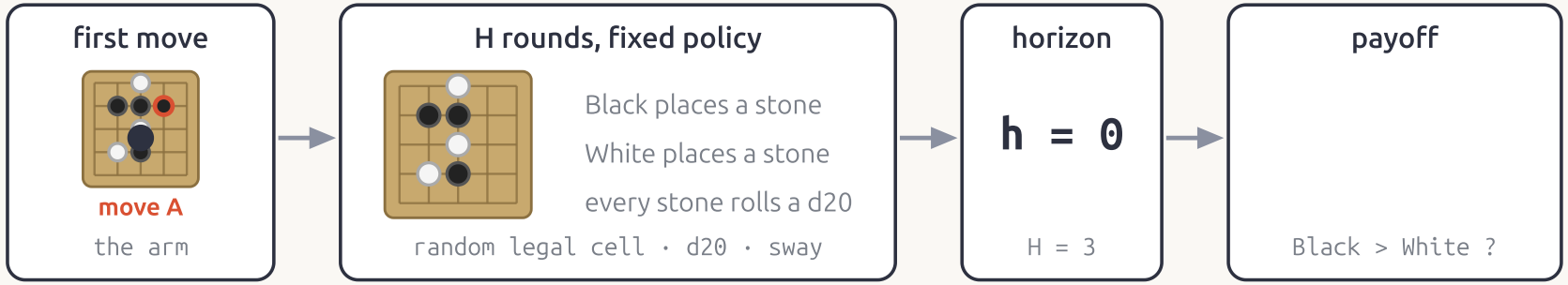
Sway benchmark







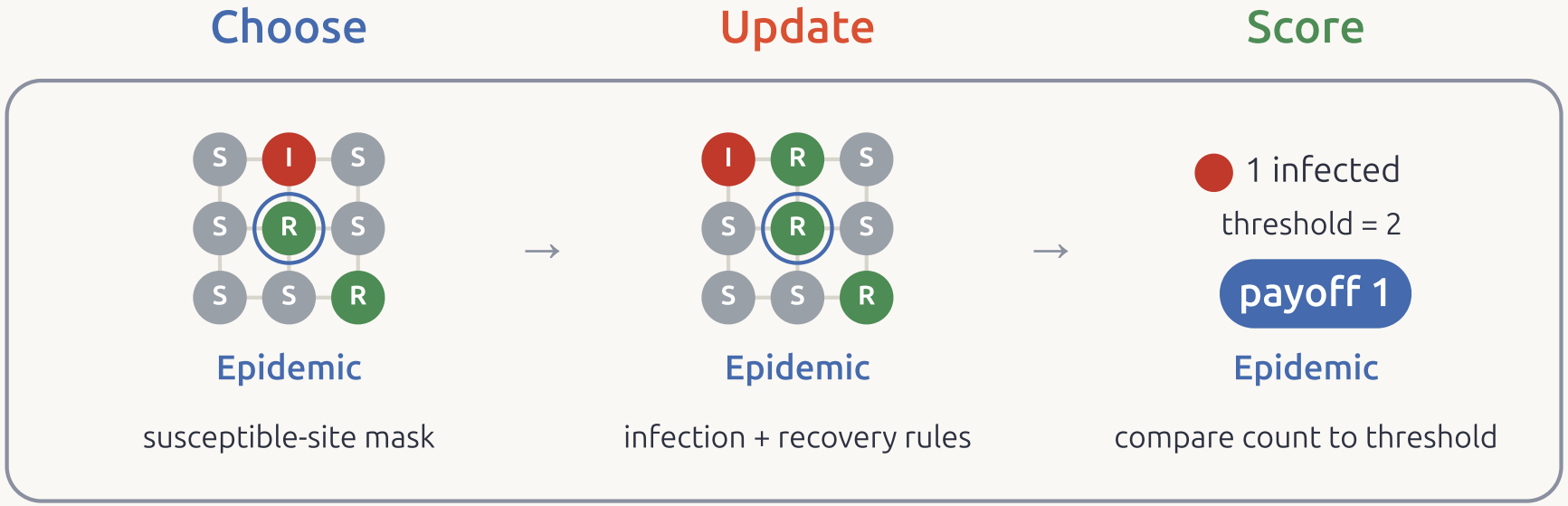
Follow one Sway rollout



running average: –

$$v_i = \Pr\left[\text{Black outnumbers White at } H \;\middle|\; \text{first move } i,\ \text{fixed policy}\right]$$

What carries over to the epidemic model?



Shared reversible circuit

randomness registers · reversible updates · cleared scratch

Back 4 / 4 · Swap the score Complete

The structure carries over. State encoding and gate costs depend on the model.

Oracle correctness check

Domain	Instance	Qubits	Depth	Gates	MC Payoff	Exact
Sway	$3\times3, H=2$	169	3 079	9 768	$.281 \pm .028$	$.271$
Sway	$5\times5, H=3$	667	18 258	55 597	$.337 \pm .021$	$.325$
Epi.	$3\times3, H=2, T=2$	146	2 383	6 472	$.883 \pm .020$	$.891$

Direct compiled resource counts

m	H	N	**Qubits, Sway**	**Qubits, Epi.**	**Gates, Sway**	**Gates, Epi.**
5	5	25	916	767	76 720	56 602
7	5	49	1 708	1 452	180 615	124 735
10	5	100	3 363	2 893	481 201	280 230
10	10	100	6 503	5 463	793 901	558 995
20	10	400	25 189	21 409	6 072 641	2 592 183

Pre-decomposition, before fault-tolerant overhead

Conclusion

- Reversible rollout oracle: $\widetilde{O}(\sqrt{k}/\varepsilon)$ calls, $O(w)$ clean ancillae
- Classical floor $\Omega(k/\varepsilon^2)$, at every member of an exponential family
- Coherent rank-select: scan $O(Nw)$, gate-optimal under bounded span; blocked $O(N \log w)$
- Open: $\Omega(N)$ vs $O(N \log w)$; lifting beyond $\kappa p < 1$; closed-loop policies